

Raj Shah

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WORK EXPERIENCE

BlackCloak AI - Remote

Jan 2026 – Present

AI Engineer

- Designed event-driven automation workflows using pub/sub architecture to triage customer security events.

Aires – Pittsburgh, PA

May 2025 – Dec 2025

Data Scientist

- Modeled time-series forecasts of different cash flow processes using foundation forecasting models.
- Automated auditing and report tasks using LangGraph and Bedrock, reducing operational tasks time by 85%.
- Developed Bayesian hierarchical models to understand fixed and random effects on customer survey scores.

Allianz Life – Minneapolis, MN

May 2024 – May 2025

Data Scientist

- Cut down compliance expenses \$20,000 annually by automating legal document classification using ML models.
- Constructed tooling comparing ML (BERT, LGBM, sklearn, etc) across 100+ permutations to achieve 84% AUC-PR.
- Identified potential lapses worth \$1,000,000+ leveraging Cox's Penalized Proportional Hazard Model in SkSurv.

Ray Analytics (Startup) – Washington, DC

Dec 2022 – July 2023

Founder/Chief Technology Officer

- Developed SaaS Platform capable of on-demand political discourse analysis and policy based influence identification.
- Trained and deployed XGBoost (94% ROC-AUC) to classify 12,000+ users based on tweet patterns and metrics.
- Employed AWS API Gateway, Lambda, and React to develop automated backend service with live analysis.

Plainspoken Digital (Startup) – Washington, DC

Mar 2022 – Dec 2022

Senior Software Engineer/Project Lead

- Led migration from legacy data systems to Palantir Foundry, improving operational reporting efficiency by 2x.
- Designed automated ETL workflows (PySpark) integrating cloud and on-prem datasets, enabling unified analytics.
- Lead Agile processes for a development team of 7 and managed intra team project management.

Amazon Web Services (AWS) – Crystal City, VA

Oct 2020 – Mar 2022

Software Engineer – L5

- Lead development and design on S3 Encryption by Default, bolstering security for billions of daily requests.
- Implemented S3 Bucket Keys, reducing encryption-related cost overhead by up to 98% for enterprise customers.
- Utilized design patterns and unit testing when deploying Java on S3's Apache Tomcat Webserver.
- Coordinated deployments across 50+ internal teams, streamlining releases and standardizing GA releases.

Data Science for Social Good – Jacksonville, FL

May 2020 – Aug 2022

Data Scientist

- Lead a team of three to analyze data for 6000+ adult learners being helped by literacy services in Florida.
- Conducted causal analysis to evaluate program efficacy and identify key drivers of learner attrition.

Booz Allen Hamilton – Washington, DC

June 2019 – Aug 2019

Data Scientist

- Programmed a Convolutional Neural Network with TensorFlow with 88% Precision, utilizing transfer learning.

EDUCATION

Carnegie Mellon University, Heinz College – Pittsburgh, PA

Master of Science, Data Science and Public Policy

Graduating May 2025

- **Relevant Coursework:** Immediate Econ Analysis, Data Analysis, Machine Learning, Applied Econometrics, Telling Stories with Data, Optimization, Introduction to Deep Learning, Voting Disenfranchisement, Decision Analytics, Geographic Information Systems, Sports Analytics, Labor Economics, Unstructured Analysis

University of Florida, College of Engineering – Gainesville, FL

Bachelor of Science, Computer Science, Minor in Mathematics

Graduated May 2020

- **Relevant Coursework:** Differential Equations, Calculus 3, Linear Algebra, Digital Logic, Algorithmic Design, Numerical Analysis, Data Structures, Digital Logic, Introduction to Data Science, Database Systems

SELECTED PROJECTS

Contracting Work – Surge HQ

Oct 2025 – Jan 2026

AI Systems and Infrastructure Setup

- Developed full-stack system for Creator-Brand Management dashboard with automated social media AI analysis.
- Deployed Cloud Run Functions that ran multiple ML models, analyzing videos using Whisper, YoloV8, OpenCV, etc.
- Created back-end that allowed brands to monitor mentions and engagement metrics across their networks.

Master's Degree Capstone Project

Jan 2025 – Apr 2025

Outcome Evaluation of Pittsburgh Employment Re-Entry Services

- Led causal inference on 2,800+ to evaluate impact of services on employment, earnings, and recidivism.
- Designed quasi-experiments and causal inference frameworks to measure program impact using social data.
- Quantified program impact on employment, calculating 3x higher odds of employments for matched graduates.
- Built panel datasets spanning 15+ years, engineering time-variant features to evaluate outcomes in Python.

Sports Analytics

Jan 2025 – Apr 2025

What's Driving the Decline in Batting Averages in MLB?

- Developed Bayesian fixed-effects model using 240,000+ pitches to predict hit prob. for causal trend analysis.
- Used brms in R to implement Stan models and visualize pitch-level MLB data with reproducible outputs.
- Modeled random effects for batter/pitcher/year/team to partition variance in hitting outcomes.
- Quantified impact of factors on hitting to conclude players impact variance more than year-to-year trends.

Introduction to Deep Learning

Aug 2024 – Dec 2024

Achieving Group and Individual Fairness in Federated Learning

- Implemented novel approach of improving Individual Fairness in Federated Instances using Flower framework.
- Trained Neural Network to classify recidivism cases from COMPAS dataset without reduction in Accuracy.

Decision Analytics and Business Policy

Aug 2024 – Dec 2024

Fund the Vote: Reducing Allegheny Voter Wait Times using Predict-then-Optimize

- Formulated framework to predict local voting turnout and optimize precinct resources to reduce wait time.
- Utilized Census data, GeoPandas, and LassoCV with a hierarchical Regression model to predict Voter Turnout.
- Developed Optimization Model using Gurobi to calculate voter wait times and effect on marginalized voters.

Geographic Information Systems

Aug 2024 – Dec 2024

Changes in Industry in Pittsburgh

- Created an ArcGIS Dashboard showing changes in occupation makeup across several years using Selectors.
- Performed Multivariate Clustering across all years to identify + visualize patterns across Pittsburgh precincts.

Telling Stories with Data

Mar 2024 – May 2024

Disproving the NBA Contract Year Phenomenon

- Utilized Scrollama, D3.JS, and React to compare player performance metrics in relation to free agency status.
- Scraped several datasets and feature engineered data using statsmodel to compare player distributions.

SKILLS

Languages & Tools: Python (Pandas, Scikit-Learn, PyTorch, statsmodels), SQL, R (Stan), Java, C++

Cloud & Infrastructure: AWS (Lambda, S3, API Gateway), GCP (Cloud Run, Pub/Sub), Render, Railways, Supabase

Modeling & Statistics: Bayesian Hierarchical Models, Causal Inference (RCTs, Matching), Forecasting, Econometrics

Visualization & BI: Tableau, GIS (ArcGIS, GeoPandas),

Optimization & Analytics: Gurobi, Metric & KPI Design, A/B Testing